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General Education Reflection: Three Questions

During my undergraduate studies, my non-technical courses, microeconomics and sociology, have influenced my perspective on the world. These courses' influence stems from the fact that they exposed me to questions that have implications for both engineering practice and everyday life.

Through my microeconomics course, I was taught that, for a business in a functioning economy, profits and losses are messages signaling either that a product is desirable or that a product needs change. The second key principle I was taught is that I should look skeptically at claims of benefits with no apparent trade-offs. Goods are only free if there is an unlimited supply, but unlimited supplies do not exist; thus, nothing is truly free. When considering the cost of goods, the key is to always consider the true cost of goods regardless of their cost to oneself. While merely a sampling of the issues covered in microeconomics, the two principles I have mentioned here are readily applicable to engineering practice: when making engineering design decisions, it is important that engineers carefully weigh their users' needs; if they do not target the needs correctly, the costs will tell them they have missed, and one should be ready to find the hidden costs in a solution that, on first blush, appears too good to be true. For instance, asking questions like an economist is especially relevant to determining whether to use AI to assist an engineering task. For example, would using AI lead to a solution that is difficult to debug once implemented? Would heavy reliance on AI in a particular capacity cause an atrophy of engineering judgment, resulting in a design the engineer does not diligently assess? In this way, economists' questions about markets can apply to cost-benefit analysis in engineering practice.

Rather than looking at how scarce resources are allocated, sociology seeks to understand the origins and workings of social interaction, whether that be large-scale trends, like increasing divorce rates, or local cultural influences, like whether children share their parents' political views. While an economist analyzes what the market demands, sociologists ask what societal trends drive this demand. Thinking like a sociologist is important in engineering as well; it helps inform ethical decision-making. When designing a product that targets a consumer market, engineers should carefully consider how the device they are creating might influence society and influence local cultures. As a clear example of this phenomenon, the introduction of the smartphone in 2007 has already had an outsized influence on society and how people communicate and interact. The design of AI tools is a current engineering problem on which engineers have direct influence. The engineers designing these AI tools should carefully weigh whether the behavior of their AI tool will lead to a more disconnected, and thus potentially more volatile society. If the engineer judges these risks as serious, they should then identify ways in which they can design their AI tool to avoid this issue. This may mean that the engineer must disable certain behaviors that could otherwise cause

individuals to seek interactions with AI when they should be interacting with other people. Whatever product an engineer designs, whether that be smartphones or AI models, they must consider how their product may positively or negatively impact social behavior regardless of how neutral their intentions may be. Thus, the exploratory questions that sociologists ask enable engineers to consider the critical questions that ensure their decisions are grounded in a desire to ensure society continues to change for the better.

Together, the viewpoints brought forth by economics and sociology form a framework of perspectives that is directly applicable to professional engineering practice. As an engineer, I should strive to consider my work from an economic and sociological perspective. An engineer's work must always involve an economic consideration: what is the optimal set of features and specifications that will meet the required market demands? Engineers should reflect on how the product they make affects society: Does it improve social outcomes, or will its non-obvious effects harm society over the long term? Taken together, the perspectives provided by microeconomics and sociology provide two key perspectives through which the various aspects of an engineer's work should be viewed.